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Robust Residual Correction via Selective Deployment for Time Series Forecasting

(/pdf?id=C5U25I5APG)

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Revisions (/revisions?id=C5U25I5APG)
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TL;DR: We propose CRC, a plug-and-play framework that deploys post-hoc residual corrections only when validation evidence indicates net error reduction over a frozen baseline forecaster.

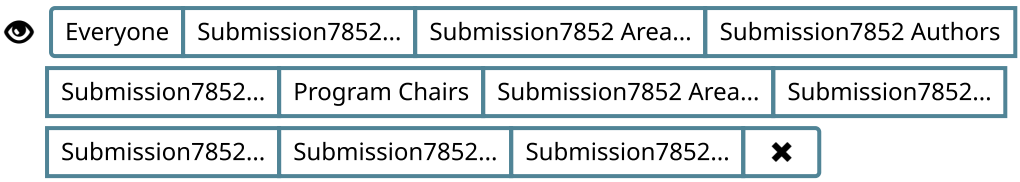
Abstract:
Post-hoc residual correction can reduce forecasting error, but always-on updates often fail to generalize: when residual signals are weak or noisy, expressive correctors can overfit and even increase the overall forecasting errors. We introduce CRC, a robust residual correction framework that couples a hybrid corrector with a validation-calibrated selective deployment rule. CRC proposes corrections via a conservative ridge "floor" plus conditional nonlinear refinement, but deploys them only when held-out evidence indicates net error reduction over a frozen baseline forecaster. This reframes correction as a decision problem: when to revise a forecast, not only how to fit residuals. Across standard long-sequence benchmarks and multiple backbones, CRC improves horizon-averaged MSE and MAE in most dataset-backbone pairs, with particularly large gains on complex datasets (e.g., over 20% relative MSE reduction on Traffic). We additionally report the non-degradation rate (NDR) as a stability diagnostic of harmful updates and ablate the validation-calibrated "firewall" that stabilizes MSE gains. Formal analysis motivating the design appears in Appendix A under idealized assumptions.

Checklist Confirmation: I confirm that I have included a paper checklist in the paper PDF.
Responsible Reviewing: We acknowledge the responsible reviewing obligations as authors.
Primary Area: General machine learning (e.g., core contributions in supervised and unsupervised methods)
Secondary Area: Deep learning advancements (e.g., architectures, optimizers, representation learning)
Contribution Type: General: Most submissions will fall into this type.
Academic Integrity: I acknowledge that I have read the NeurIPS Handbook and commit to adhering to all policies in the Handbook (https://neurips.cc/Conferences/2026/MainTrackHandbook (https://neurips.cc/Conferences/2026/MainTrackHandbook)), the NeurIPS Code of Conduct and the NeurIPS Academic Integrity Policy.
LLM Usage: Understanding technical concepts
Declaration: I confirm that the above information is accurate.
Reviewer Nomination: YUNCHENG HUA (/profile?id=~YUNCHENG_HUA2)
Submission Number: 7852

Discussion (?id=C5U25I5APG#discussion)

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Meta Review of Submission7852 by Area Chair asfz

Meta Review by Area Chair asfz 21 Jul 2026, 20:27 (modified: 24 Jul 2026, 03:30)

Senior Area Chairs, Area Chairs, Authors, Reviewers Submitted, Program Chairs, Area Chair asfz

Revisions (/revisions?id=dec225z0lL)

Metareview:

This submission proposes CRC, a plug-in residual corrector for frozen forecasting backbones that decomposes candidate corrections into a ridge "floor" plus a clipped, direction-gated MLP refinement, and its central idea deploys corrections selectively. A per-(variable,horizon) choice among ,baseline, +linear, and +linear+MLP is calibrated on validation and frozen at test time.

Strengths (consensus of reviewers):

- the "when to deploy, not only how to fit" framing is well motivated and practically relevant. The evaluation is broad. NDR is a useful diagnostic that averages obscure. Component ablations and sensitivity analyses (tau rescaling, validation budget) support the configuration. The paper is clear, honest about limitations, and code is released.

Weaknesses (summarized and augmented by the ACs question below):

- Uncertainty about novelty and baselines: See ACs follow-up question below.
- Missing statistics: No variability such as seed variance is reported despite averaging (raised by 14dv; vaUX/gnqC relatedly). Many Table 1 gains are quite small and the strongest-backbone (TimeXer) column includes two negative entries, so the "26 of 28 pairs" claim is currently weak without a more formal statistical analysis.
- Missing controls: The "always-on" control is missing (vaUX, gnqC): Table 2 removes filters one at a time but never shows the full corrector with no firewall, nor a simple validation-gated ridge, so the gain attributable to selective deployment itself is not isolated.
- Dependence on validation-test representativeness (MkCF, gnqC) deserves a discussion.

In summary, the paper has merit and addresses an important problem with the reviews being borderline accept and positive; however, there are currently non-trivial question/concerns that needs addressing before it can be considered for acceptance. The authors are encouraged to address the issues summarized above along with and the individual questions posed by the reviewers and follow up question included by the AC (below). In particular, if any of the reviewers' or AC's reservations are factually wrong, the authors should say so in the rebuttal.

AC follow-up questions to ensure an informative rebutal:

- AC-Q1: This is a very crowded area, and the novelty clarification needs a bit of clarification as far as I can tell. The paper seems to miss a few papers in the related work section, and possibly even as baselines, that appear highly relevant including TAFAS (AAAI 2025) which applies gated residual correction to frozen forecasters' per variable. PETSA and FAC use learnable per variable gates. SOLID adapts only when a residual-based detector warrants it. PIR's (cited) identification stage already modulates revision strength per instance as far as I can decode from the paper, so Section 4.3's descriptin of the different to PIR is not a big as descriped. Per-unit holdout selection among candidate forecasts is also classical (FFORMS/FFORMA).

TAFAS: Battling the Non-stationarity in Time Series Forecasting via Test-time Adaptation"

SOLID: "Calibration of Time-Series Forecasting: Detecting and Adapting Context-Driven Distribution Shift", https://arxiv.org/abs/2310.14838 (https://arxiv.org/abs/2310.14838)

PETSA: "Accurate Parameter-Efficient Test-Time Adaptation for Time Series Forecasting", https://arxiv.org/abs/2506.23424 (https://arxiv.org/abs/2506.23424)

FAC: "Towards Principled Test-Time Adaptation for Time Series Forecasting", 2026.

Based on these papers it seems like every part of CRC exists separately in prior work (gated correction of frozen forecasters, per-unit holdout selection a la FFORMA, non-degradation as a negative flip metric). Could the authors please help the AC by commenting on these papers and concisely explain what the novelty of the specific synthesis is? Also, it would be usefull to learn why the above methods have not been incldued as baselines.

- AC-Q2: Appendix A seems to oversell in that Prop A.3 is true by construction as far as I can see, and Prop A.4's displayed bound does not follow as stated (remark A.5 admits this), i.e., the abstract's "formal analysis" phrasing seems like a bit of a stretch. Could the authors comment, please?

Add: **Author AC Confidential Comment**

Official Review of Submission7852 by Reviewer vaUX

Official Review by Reviewer vaUX 24 Jun 2026, 11:19 (modified: 23 Jul 2026, 21:54)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer vaUX

Revisions (/revisions?id=8T1RvdEK8G)

Summary:
The paper explores a situation in time series forecasting when an update to a model is being deployed and comments on the fact that the default mode of "always-on" update (so always accept the update) may lead to several problems. They propose a solution which is based on the fact that there is a set aside held-out set which can be consulted as to see if the new update would cause decrease in performance. Depending on this they effectively reframe the problem as a decision problem of "when to deploy". The results show that their method brings some improvements. There are however some limitations in the paper which are described in more detail below.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:
Strengths

Motivation for the paper is good and well written. This sounds like an interesting problem and by reframing it to a "when to deploy" task sounds novel albeit perhaps simple (but if its elegantly simple and it works in this new framing that's a strenght on its own as well)

Weaknesses

Background

Background could be significantly expanded and better explained. There is not enough information given about the tasks of the datasets and also about the backbones. It would help the paper to feel less abstract, if the data and tasks solved by these methods would be better explained and illustrated. Additionally, when the description of these datasets is missing, one wonders about what happens with the inevitable real world anomalies (OOD data, sensor noise, extreme events) and this method - do the used benchmarks capture these or not? This is also linked to the language used - e.g. How to we "learn a correction" (does that mean that model is finetuned on more data?)? How do we "measure the improvement" (does that mean that we measure performance metrics on set aside data?)?

Results / simple baseline

In Table 1 and throughout the text a potentially simple baseline is missing - "always accept update". It is described in the motivation of the paper (when defining the identified gap), but then the paper itself never really compares against that simple approach. Unless this is what is meant by the "baseline" in Table 1, in which case the other simple approach (of "never update") is missing.

Results / limited comparisons

The NDR result give more insight into the results (over just the MSE numbers and % improvements), but they are missing in some places (e.g. Table 1).

Missing details

Not enough details are given about the held out calibration set. How was it made? Is it representative? Is it difficult to make (thus adding cost to using CRC?)?

Quality: 3: good
Clarity: 3: good
Significance: 2: not good
Originality: 3: good
Questions:
Major suggestion:

- Add missing baselines ("always update", "never update" vs the proposed "update when it's smart" which is your CRC method), or explain why they wouldn't make sense (if that's the case). Otherwise the claims made throughout the whole paper are not really founded on real evidence.

Minor points:

- Improve the background description (as in expanded on in the weaknesses). Also, while you briefly mention what PIR is in the Section 2, consider adding more details about how this method works here.
- Suggestion to restructure the paper so that there is a section called Results (right now its Experiments and Discussion directly). Maybe split it after 4.2. "Main result" and the following into it's own Results section

Limitations:
yes

Rating: 4: Borderline accept: Technically solid paper where reasons to accept outweigh reasons to reject, e.g., limited evaluation. Please use sparingly.
Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
None

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes

Add: **Rebuttal** **Author AC Confidential Comment**

Official Review of Submission7852 by Reviewer 14dv

Official Review by Reviewer 14dv 19 Jun 2026, 23:17 (modified: 23 Jul 2026, 21:54)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 14dv
Revisions (/revisions?id=NBJAk0L08s)

Summary:
This paper reframes post-hoc residual correction as a decision problem: not only how to fit residuals, but when a forecast should be revised at all. The standard approach stacks an auxiliary module on a frozen backbone to correct its residual error, applying the correction unconditionally at test time. The authors argue this always-on regime is fragile, since a correction that helps in aggregate can worsen the error in individual cases. They call this the corrector's dilemma: biases that are reducible on average must be shielded from noise amplification at deployment.

CRC takes the history and the baseline prediction as input and selects between a linear or both a non-linear and linear refinement path using held-out evidence. The data is segmented with a sliding window, then two branches produce a rich embedding Z (one extracting statistics like means, trends, and correlations, the other creating averages over pairwise embeddings based on correlation-based K-nearest neighbors over variables). A residual estimator decomposes the correction into a linear part (ridge) and a non-linear part (MLP), with the non-linear

term added conditionally under a deployment policy that targets failure modes like sign flips and selection noise. The policy accepts non-linear refinement only when its sign matches the linear correction, clips corrections with validation-calibrated thresholds, freezes each scalar correction to linear-only or hybrid based on validation, and deploys only when validation gains exceed a margin.

Contribution Type: General: Most submissions will fall into this type.
Strengths And Weaknesses:
The paper addresses an interesting and well-motivated problem, and treating correction as a conditional decision rather than an unconditional transformation is a useful reframing. The presentation is a clear strength, the structure is easy to follow, they provide a formal analysis, and the experiments are seeded. The architecture is properly ablated, and the NDR gains attributable to each component are convincing, making a strong case for the design choices. The evaluation is also broad, spanning multiple standard benchmarks and standard backbone architectures.

The main weaknesses are reporting-related rather than conceptual. Variability across seeds is not surfaced, and the statistical significance of the improvements over the baseline is not established (see Questions). Neither gap undermines the contribution, but addressing both would strengthen confidence in the reported gains.

Quality: 3: good
Clarity: 4: excellent
Significance: 3: good
Originality: 3: good
Questions:
- Can you add the variability across seeds?
- Are the differences between PIR and CRC statistically significant (Table 3)?

Limitations:
None

Rating: 5: Accept: Technically solid paper, with high potential value on at least one sub-area of AI or moderate-to-high impact on more than one area of AI, with good-to-excellent evaluation, resources, reproducibility, and no unaddressed ethical considerations.
Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
None

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes

Add: **Rebuttal** **Author AC Confidential Comment**

Official Review of Submission7852 by Reviewer MkCF

Official Review by Reviewer MkCF 18 Jun 2026, 00:24 (modified: 23 Jul 2026, 21:54)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer MkCF
Revisions (/revisions?id=mFrolHwJNr)

Summary:
This paper investigates the post-processing residual correction problem in time series forecasting. Traditional methods typically train a residual corrector after freezing the backbone forecasting model to correct prediction bias. However, existing methods often employ always-on correction, directly updating residuals regardless of whether correction is needed. This can easily lead to overfitting when residual signals are weak or noise is high, increasing test errors. To address this, this paper proposes CRC, transforming residual correction from a question of "how to fit the residuals" to "when to deploy the correction." CRC first generates candidate residuals using a hybrid corrector, then uses a selective deployment rule calibrated on a validation set to determine whether to apply the correction, thereby reducing harmful updates and improving prediction stability.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths:

1.The authors changed the focus of posterior residual correction from "how to fit the residuals" to "when to deploy the correction." This perspective is more reliable and better aligns with the stability requirements of practical applications than always-on correction.

2.The authors propose a CRC framework that combines direction-aware residual representation, ridge floor, conditional MLP refinement, and validation-calibrated deployment firewall to achieve more effective residual correction.

3.The authors propose using NDR as an auxiliary diagnostic indicator to compensate for the problem that MSE/MAE can only reflect average performance and is difficult to expose local degradation.

4.The paper validates its core argument through comparison with PIR and multi-component ablation: the performance gains of residual correction come not only from a stronger corrector, but also from the selective deployment rules of validation set calibration.

Weaknesses:

1.The CRC deployment rules are calibrated on the validation set and frozen into the testing phase. The problem is that this set of rules may fail if the validation and test sets are not distributed correctly.

2.CRC uses a KNN graph based on the correlation of training data to construct variable adjacency relationships, with a default K=5, and then uses this graph consistently during the validation and testing phases. This graph can only capture simple correlations and may not be able to represent nonlinear relationships, time-delay relationships, or dynamically changing relationships.

3.The experiments covered 7 long sequence prediction datasets and 4 backbones, but mainly standard benchmarks such as ETTh, ETTm, Weather, Electricity, and Traffic. I think the focus should be on testing issues such as online updates and strong distribution drift.

4.There are many hyperparameters and rules, and the migration cost is not low.

Quality: 3: good
Clarity: 3: good
Significance: 3: good
Originality: 3: good
Questions:
1.How often do deployment rules need to be recalibrated?

2.What are the criteria for choosing K=5 in KNN?

3.What is the relationship between NDR and MSE/MAE?

Limitations:
The author did not adequately discuss the limitations he used.

Rating: 4: Borderline accept: Technically solid paper where reasons to accept outweigh reasons to reject, e.g., limited evaluation. Please use sparingly.
Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
N/A

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes

Add: **Rebuttal** **Author AC Confidential Comment**

Official Review of Submission7852 by Reviewer gnqC

Official Review by Reviewer gnqC 08 Jun 2026, 16:15 (modified: 23 Jul 2026, 21:54)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer gnqC

Revisions (/revisions?id=o7bvUApILc)

Summary:
This paper introduces a post-hoc residual correction method for time series forecasting. Based on the observations that always-on correction methods can improve some predictions but hurt others, the paper's main idea idea is to avoid applying residual corrections everywhere. The proposed CRC combines a ridge-based correction, a lightweight nonlinear refinement, and a validation-calibrated deployment rule to decide where corrections should be used. The method is evaluated on several standard long-horizon forecasting datasets with multiple backbone models. The results show improvements in MSE/MAE in most settings, with additional analysis using non-degradation rate (NDR) to measure how often corrections avoid worsening local forecasts.

Contribution Type: General: Most submissions will fall into this type.
Strengths And Weaknesses:
Strengths

1.The paper addresses a practical and important problem. The post-hoc correction can be useful, but unsafe corrections may degrade already good forecasts. 2.The selective deployment presents a novel perspective. Treating correction as a "when to deploy" problem, rather than only a residual fitting problem, is an intuitive and useful framing. 3.The experiments cover several standard datasets and backbone models, and the comparison with PIR is a useful addition. 4.The ablation study helps show the effect of the ridge floor, nonlinear refinement, encoder, and deployment firewall. 5.The NDR is useful, since aggregate MSE/MAE can hide local degradations across variables and horizons. 6.The paper is generally clear and easy to follow. The codes are also available.

Weaknesses

1. CRC has several moving parts. The paper would benefit from a clearer distinction between the core selective deployment mechanism and stabilizing details such as clipping, direction gating, point-wise selection, and shrink-to-base blending.
2. Since validation-calibrated deployment is central to CRC, a comparison with simpler validation-gated baselines would strengthen the empirical story. Without this comparison, it is somewhat hard to tell how much of the gain comes from the full CRC design rather than validation-based selection alone.
3. CRC may depend on how representative the validation period is. A clearer discussion of validation-test mismatch and possible recalibration would make the scope more transparent.
4. NDR is useful, but its relationship with MSE/MAE could be explained more clearly, since high NDR does not necessarily imply the largest aggregate error reduction.

Quality: 3: good
Clarity: 3: good
Significance: 3: good
Originality: 3: good
Questions:
1. Since validation-calibrated deployment is a key part of CRC, I would like to better understand how much of the gain comes from the full CRC design rather than from validation-based selection alone.
2. I find NDR useful, but I was not fully sure how it should be interpreted together with MSE/MAE. For example, in what cases should a practitioner prefer a method with higher NDR but slightly weaker average error?
3. The validation-ratio sensitivity analysis is helpful. Could the authors also comment on what may happen when the validation period is not very representative of the test period? A short discussion of possible failure cases or recalibration would be useful.
4. CRC has a few components, and I think the paper would be easier to read if the authors more clearly separated the core idea from implementation details. In particular, which parts are essential to the selective deployment mechanism, and which parts are stabilizers that could potentially be replaced?

Overall, I view the paper positively. My main questions are about the source of improvement and the scope of the full CRC design.

Limitations:
Mostly yes. The paper discusses relevant limitations, including the fixed graph prior, conservative gating, and dependence on the validation period. I would suggest making the validation-test mismatch point slightly more prominent, since the deployment rule is central to CRC.

Rating: 5: Accept: Technically solid paper, with high potential value on at least one sub-area of AI or moderate-to-high impact on more than one area of AI, with good-to-excellent evaluation, resources, reproducibility, and no unaddressed ethical considerations.

Confidence: 5: You are absolutely certain about your assessment. You are very familiar with the related work and checked the math/other details carefully.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

No problem

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

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